Domain Shift Analysis via Overtrained Neural Networks to Guide Source Dataset Selection in Cross-Domain Few-Shot Object Detection

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Abstract

Domain shift is a challenge in machine learning, arising when the statistical distribution of training data (source domain) diverges from that of target data (target domain). In aerial image object detection, domain shift is particularly noticeable when acquisition conditions, geographic contexts, sensor types, and annotation processes differ significantly across datasets. In this work, we propose a systematic framework to *quantify* domain shift between multiple aerial and general-purpose image datasets by leveraging **overtrained neural networks** as domain discriminators within a **few-shot learning** paradigm. Embedding representations are extracted using a pretrained visual encoder and subsequently used to train lightweight classifiers, including Logistic Regression, Multilayer Perceptrons (MLP), and dedicated Domain Discriminators, to distinguish between pairs of domains under varying numbers of labeled examples (shots). Domain shift is then quantified through a suite of complementary metrics: the *Domain Separability Index* (DSI), Area Under the learning Curve (AUC), convergence rate, prediction stability, and classical distributional distances (Wasserstein-1 and Multi-Kernel Maximum Mean Discrepancy). We apply this framework to thirteen datasets spanning aerial imagery, everyday objects, fashion, food, and underwater domains, identifying which source datasets exhibit the lowest shift with respect to canonical aerial target datasets (DOTA and DIOR). Our results provide actionable guidance for source dataset selection in cross-domain few-shot object detection pipelines.

Keywords: domain shift, few-shot learning, object detection, aerial imagery, domain discriminator, Wasserstein distance, MMD, embedding space, source dataset selection.

1 Introduction

1.1 Problem Statement

In machine learning, **domain shift** occurs when the data used to train a model differs from the data it encounters during inference. When a model trained on one dataset is applied to another, even for the same task, its performance frequently degrades significantly. Understanding and *quantifying* this shift is therefore critical to building robust, generalizable systems.

This study focuses on **aerial image datasets** used for few shot object detection, a domain that presents a particularly challenging instance of the domain shift problem due to the following factors:

- **Acquisition condition variability:** altitude, image resolution, and field of view differ significantly across datasets.
- **Geographic context:** datasets captured over urban environments, rural landscapes, or exhibit structurally different scene layouts.
- **Annotation protocol discrepancies:** category definitions, bounding box granularity, and labeling conventions are not standardized across the community.
- **Image quality degradations:** blur caused by motion or atmospheric turbulence, non-uniform lighting further complicate cross-dataset generalization.

1.2 The Few-Shot Framework for Domain Shift Measurement

We evaluate domain shift through a **few-shot learning** lens: rather than training on thousands of labeled examples, we ask : *how quickly can a classifier learn to distinguish between two domains when given only a small number of examples (shots) per class?*

This approach is based on two complementary insights:

1. **Low separability** between two domains indicates *small domain shift*, meaning that features transfer well from source to target.
2. **High separability even with few shots** suggest *large domain shift*, indicating that the two datasets occupy fundamentally different regions in the embedding space.

Therefore, the few-shot setting provides a practical and data-efficient way to examine the structure of the embedding space. This allows for analysis of domain shifts, even when large amounts of labeled data are not available.

1.3 Contributions

The main contributions of this work are as follows:

- We design a **reproducible pipeline** for computing, storing, and comparing embedding representations of large-scale image datasets using streaming data loading.
- We propose a **composite Domain Shift Index (DSI)** that aggregates few-shot learning curves, convergence metrics, and distributional distances into a single interpretable score.
- We perform a **large-scale empirical study** across thirteen heterogeneous image datasets, providing a ranked ordering of source datasets by their proximity to two aerial target benchmarks: DOTA and DIOR.

- We provide **actionable guidelines** for practitioners selecting source datasets for cross-domain few-shot object detection.

2 Related Work

2.1 Domain Adaptation and Domain Shift

Domain adaptation has been extensively studied in computer vision. Classical approaches include feature alignment via Maximum Mean Discrepancy [6], adversarial domain adaptation [5], and optimal transport-based methods [3]. These methods generally assume access to both source and target data simultaneously, whereas our setting focuses purely on *measuring* shift without necessarily adapting the model.

Distributional distance measures such as the **Wasserstein distance** [12] and **Maximum Mean Discrepancy (MMD)** [6] have become standard tools for quantifying shift between feature distributions. In high-dimensional settings, however, these metrics suffer from the curse of dimensionality, motivating the use of dimensionality reduction (e.g., PCA) prior to distance computation—a strategy we adopt in this work.

2.2 Few-Shot Learning

Few-shot learning seeks to generalize from a limited number of labeled examples [13, 4]. In the context of domain shift measurement, few-shot binary classification between two domains serves as a *discriminability probe*: the speed at which a classifier achieves high accuracy directly reflects the magnitude of the distributional gap.

Theoretical grounding. This hypothesis is formally rooted in the $\mathcal{H}$-divergence framework of Ben-David et al. [2], which bounds the target error of any hypothesis $h \in \mathcal{H}$ as:

$$\epsilon_T(h) \;\leq\; \epsilon_S(h) \;+\; \frac{1}{2}\, d_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) \;+\; \lambda^*, \tag{1}$$

where $\lambda^* = \min_{h\in\mathcal{H}}[\epsilon_S(h) + \epsilon_T(h)]$ is the ideal joint error. Crucially, the $\mathcal{H}$-divergence admits an empirical estimator via a binary domain classifier:

$$\hat{d}_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) \;=\; 2\left(1 - 2\inf_{h\in\mathcal{H}} \hat{\epsilon}(h)\right), \tag{2}$$

where $\hat{\epsilon}(h)$ is the classification error of h when trained to discriminate source from target samples. This quantity is known as the **Proxy $\mathcal{H}$-Divergence (PAD)** [5]. A classifier achieving near-perfect accuracy ($\hat{\epsilon} \to 0$) implies PAD $\to 2$, signaling maximal distributional divergence, whereas chance-level accuracy ($\hat{\epsilon} \to 0.5$) implies PAD $\to 0$, indicating near-identical distributions.

The few-shot regime as a structured probe. Rather than training the domain classifier on the full dataset, we deliberately restrict training to $k \in \{1, 5, 10, 25, 50, 100, 250, 500\}$ labeled examples per class, tracing a complete *learning curve*. This design is motivated by two complementary observations. First, Vinyals et al. [13] and Snell et al. [10] demonstrate that even with $k = 1$ or $k = 5$ examples, classifiers operating on rich embedding spaces reliably capture the geometric separation between class distributions. Second, Ravi and Larochelle [9] establish that the *rate of accuracy gain* as a function of training set size is governed by the curvature of the loss landscape in embedding space, which in turn reflects inter-domain geometric separation. Concretely, if domains $\mathcal{D}_S$ and $\mathcal{D}_T$ are well-separated in feature space, even a single shot per class is sufficient to correctly orient a decision boundary, yielding a steep learning curve. Conversely, heavily overlapping distributions require many more examples to achieve reliable separation, producing a shallow, slowly saturating curve.

Empirical precedent. This principle was anticipated by Torralba and Efros [11], who showed that a standard image classifier trained to identify the *source dataset* of an image achieves remarkably high accuracy across ostensibly similar vision benchmarks—a direct empirical demonstration that datasets carry strong and learnable statistical signatures. Our work extends this observation into the few-shot regime and formalizes it into a quantitative shift metric, the **Domain Separability Index (DSI)**:

$$\mathrm{DSI}(k) \;=\; \frac{1}{|\mathcal{M}|} \sum_{m\in\mathcal{M}} \overline{\mathrm{AUC}}_m(k) \;\times\; 100, \tag{3}$$

where $\mathcal{M} = \{\text{LogReg, MLP, Discriminator}\}$ is the set of classifiers and $\overline{\mathrm{AUC}}_m(k)$ is the mean AUC over N episodes at k shots. A high $\mathrm{DSI}(k)$ for small k therefore constitutes evidence of large domain shift, consistent with the distributional metrics of Gretton et al. [6] and Villani [12], whose correlation with DSI is empirically validated in Section 6. Furthermore, Long et al. [8] provide evidence that discriminator accuracy and MMD are correlated in deep feature spaces, reinforcing the coherence of our multi-metric framework. Finally, Zhao et al. [16] strengthen the theoretical link by showing that low inter-domain separability in feature space is a *necessary condition* for successful transfer, directly motivating the use of DSI as a principled source dataset selection criterion in few-shot object detection [14].

2.3 Aerial Image Datasets

Benchmark datasets for aerial object detection include **DOTA** [15] and **DIOR** [7], both of which feature diverse object categories captured from satellite or drone imagery. Cross-domain generalization to these benchmarks from datasets collected under different conditions remains an open challenge.

3 Datasets

Table 1 summarizes the thirteen datasets included in this study. They span a wide range of visual domains, from overhead aerial imagery to food, fashion, and underwater scenes, providing a comprehensive testbed for domain shift analysis.

Table 1: Summary of datasets used in the study. BG variants correspond to background-only subsets.

Dataset	**Domain**	**Type**	**Notes**
DOTA	Aerial	Detection	Target benchmark
DIOR	Aerial	Detection	Target benchmark
DOTA-BG	Aerial	Background	Background patches
DIOR-BG	Aerial	Background	Background patches
COCO	General	Detection	MS-COCO
COCO-BG	General	Background	Background patches
XVIEW	Aerial/Satellite	Detection	Overhead imagery
CADOT	Aerial	Detection	Drone imagery
DEEPFRUITS	Agricultural	Detection	Fruit detection
OKTOBERFEST	Event/Food	Detection	Crowd and food
FASHIONPEDIA	Fashion	Detection	Clothing items
ARTAXOR	Biological	Detection	Insect imagery
UODD	Underwater	Detection	Underwater objects

The inclusion of semantically distant datasets (FASHIONPEDIA, OKTOBERFEST, ARTAXOR, UODD) is intentional: they serve as *negative controls*, expected to exhibit large domain shift with respect to aerial benchmarks, thereby validating the discriminative power of our metrics.

4 Methodology

4.1 Overview

Figure 1 shows the overall pipeline consists of four main stages for analyzing domain shift between datasets. First, in the embedding extraction stage, images from each dataset are passed through a pretrained visual encoder to obtain dense feature representations. Next, in the dataset pairing and balancing stage, pairs of source and target datasets are constructed, with downsampling applied to reduce large differences in dataset size.

To mimic few-shot learning scenarios, we then perform few-shot classification, where lightweight classifiers are trained on progressively larger subsets of the embeddings. This allows us to observe how easily the embeddings from different datasets can be separated under limited data conditions. Finally, domain shift metrics are computed by analyzing both the classifier learning curves and statistical distances between embedding distributions. These measurements are combined to produce a ranking of source datasets according to their similarity to the target domain.

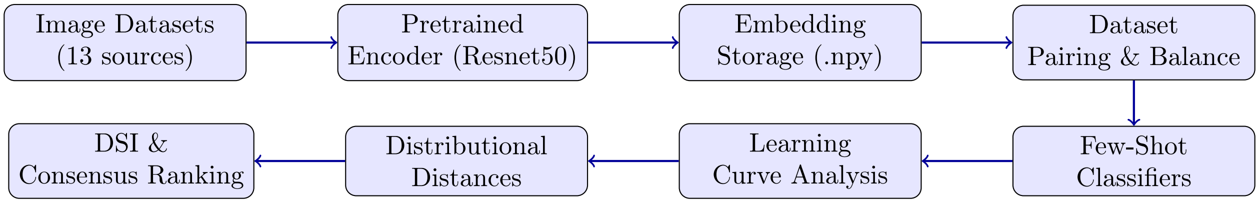


Figure 1: Overview of the domain shift analysis pipeline.

4.2 Embedding Extraction

All images are embedded using a pretrained vision encoder (Resnet50 trained on ImageNet). Embeddings are computed in streaming mode to handle large-scale datasets without loading them entirely into memory. Formally, let $f_\theta : \mathcal{X} \to \mathbb{R}^d$ denote the encoder, where $\mathcal{X}$ is the image space and d is the embedding dimension. For a dataset $\mathcal{D} = \{x_i\}_{i=1}^N$, we compute:

$$\mathbf{Z} = \{f_\theta(x_i)\}_{i=1}^N \subset \mathbb{R}^d \tag{4}$$

Embeddings are stored as `float16` arrays on disk for memory efficiency, organized by dataset. A blur pre-filtering step is applied using the Laplacian variance:

$$\text{BlurScore}(I) = \text{Var}\big(\nabla^2 I\big) \tag{5}$$

where $\nabla^2 I$ denotes the discrete Laplacian of the grayscale image I. Images with BlurScore $< \tau_{\text{blur}}$ are flagged as low-quality. This step ensures that embedding distributions are not contaminated by severely degraded samples.

4.3 Dataset Pairing and Class Balancing

For each ordered pair (A, B) of datasets, embeddings are concatenated and labeled as:

$$y_i = \begin{cases} 0 & \text{if } \mathbf{z}_i \in \mathbf{Z}_A \\ 1 & \text{if } \mathbf{z}_i \in \mathbf{Z}_B \end{cases} \tag{6}$$

To mitigate class imbalance, we apply a **downsampling** strategy that truncates both sets to size $n_{\min} = \min(|A|, |B|)$ using seeded random sampling without replacement:

$$\tilde{\mathbf{Z}}_A = \mathbf{Z}_A[\sigma_A], \quad \tilde{\mathbf{Z}}_B = \mathbf{Z}_B[\sigma_B] \tag{7}$$

where σ_A, σ_B are random permutations drawn from a fixed random seed for reproducibility.

4.4 Few-Shot Classification Protocol

For each dataset pair and each value of $k \in \{1, 5, 10, 50, 100, 200, 500\}$ shots per class, we:

1. Sample k embeddings from each domain to form the training set.
2. Train a binary classifier on this training set.
3. Evaluate on a held-out test set of fixed size.
4. Record accuracy, AUC-ROC, and the *Probability of Adversarial Discrimination* (PAD).

5. Repeat for multiple random seeds and aggregate mean and standard deviation.

Three classifier architectures are evaluated:

Logistic Regression (LogReg). A linear probe providing a lower bound on separability. Given embeddings $\mathbf{Z}$ and labels $\mathbf{y}$, it minimizes:

$$\min_{\mathbf{w},b} \sum_i \mathcal{L}_{\text{BCE}}\Big(\sigma(\mathbf{w}^\top \mathbf{z}_i + b),\, y_i\Big) + \lambda \|\mathbf{w}\|_2^2 \tag{8}$$

Multilayer Perceptron (MLP). A nonlinear classifier with architecture $d \to 128 \to 32 \to 1$, trained with Binary Cross-Entropy loss, Adam optimizer, and early stopping based on validation loss with patience $p = 5$ epochs.

Domain Discriminator. An overtrained MLP augmented with Batch Normalization layers after each hidden layer, designed to maximize discriminative power. The architecture follows the domain discriminator design principle introduced in Ganin et al. [5] and implemented in the DALIB library [17], which provides standardized, reproducible implementations of domain adaptation components. In DALIB, the domain discriminator is defined as a binary classifier trained to distinguish source from target domain embeddings, and its accuracy serves as a direct proxy for the Proxy $\mathcal{H}$-Divergence (PAD) of Ben-David et al. [2] (see Equation 21). Our implementation extends this design with an additional Batch Normalization layer to stabilize training in the few-shot regime, where gradient variance is amplified by the small number of available samples:

$$\hat{y} = \sigma(\text{Linear}_{32\to1}(\text{BN}(\text{ReLU}(\text{Linear}_{128\to32}(\cdot))))) \tag{9}$$

The full architecture is:

$$\mathbf{h}_1 = \text{ReLU}(\text{BN}(\mathbf{W}_1\mathbf{z} + \mathbf{b}_1))\,, \quad \mathbf{W}_1 \in \mathbb{R}^{128\times d}, \tag{10}$$

$$\mathbf{h}_2 = \text{ReLU}(\text{BN}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2))\,, \quad \mathbf{W}_2 \in \mathbb{R}^{32\times 128}, \tag{11}$$

$$\hat{y} = \sigma(\mathbf{W}_3\mathbf{h}_2 + b_3), \quad \mathbf{W}_3 \in \mathbb{R}^{1\times 32}, \tag{12}$$

where $\mathbf{z} \in \mathbb{R}^d$ is the input embedding, $\text{BN}(\cdot)$ denotes Batch Normalization [18], and $\sigma(\cdot)$ is the sigmoid activation. The model is trained with binary cross-entropy loss:

$$\mathcal{L}_{\text{disc}} = -\frac{1}{n}\sum_{i=1}^{n} \left[y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)\right], \tag{13}$$

where $y_i = 0$ for source domain samples and $y_i = 1$ for target domain samples. A high discriminator accuracy after few-shot training directly maps to a large PAD estimate via Equation 21, confirming significant domain shift between $\mathcal{D}_S$ and $\mathcal{D}_T$.

4.5 Distributional Distance Metrics

In parallel with the discriminative approach, we compute two distributional distances directly on the embedding sets.

4.5.1 Wasserstein-1 Distance

To mitigate the curse of dimensionality, we first apply PCA to reduce both embedding sets to $n_c = \min(50, d, |A|, |B|)$ components:

$$W_1(A, B) = \frac{1}{n_c} \sum_{j=1}^{n_c} W_1\Big(\tilde{A}_{:,j},\ \tilde{B}_{:,j}\Big) \tag{14}$$

where $W_1(p, q) = \int |F_p(x) - F_q(x)|\,\mathrm{d}x$ is the one-dimensional Wasserstein-1 distance, computable in closed form via empirical CDFs.

4.5.2 Multi-Kernel Maximum Mean Discrepancy (MK-MMD)

For a single RBF kernel $k_\gamma(\mathbf{x}, \mathbf{y}) = \exp(-\gamma\|\mathbf{x} - \mathbf{y}\|^2)$, the squared MMD is:

$$\widehat{\mathrm{MMD}}^2_\gamma(A, B) = \frac{1}{m(m-1)} \sum_{i\neq j} k_\gamma(a_i, a_j) + \frac{1}{n(n-1)} \sum_{i\neq j} k_\gamma(b_i, b_j) - \frac{2}{mn} \sum_{i,j} k_\gamma(a_i, b_j) \tag{15}$$

We use a **multi-kernel** variant averaging over 30 log-spaced values of γ, after standardizing features with a `StandardScaler`:

$$\text{MK-MMD}(A, B) = \frac{1}{|\Gamma|} \sum_{\gamma\in\Gamma} \sqrt{\max\Big(\widehat{\mathrm{MMD}}^2_\gamma(A, B),\ 0\Big)} \tag{16}$$

4.6 Domain Shift Metrics

From the few-shot learning curves $\{(k_t, \mathrm{acc}_t)\}_{t=1}^{T}$ and $\{(k_t, \mathrm{auc}_t)\}_{t=1}^{T}$, we derive the following composite metrics.

4.6.1 Domain Separability Index (DSI)

The DSI is defined as the normalized area under the AUC-vs-shots curve up to $k = 10$ shots, measuring *early separability*:

$$\mathrm{DSI}_{10} = \frac{1}{k_{\max}} \int_1^{10} \mathrm{AUC}(k)\,\mathrm{d}k \tag{17}$$

A high DSI indicates that the two domains are easily distinguishable even with very few examples, implying large domain shift.

4.6.2 AUC of the Learning Curve

The full area under the AUC learning curve is computed using the trapezoidal rule:

$$\mathrm{AUC}_{\text{learn}} = \sum_{t=1}^{T-1} \frac{(k_{t+1} - k_t)}{2} \left(\mathrm{AUC}_t + \mathrm{AUC}_{t+1}\right) \tag{18}$$

4.6.3 Convergence Rate

The convergence rate measures the speed at which accuracy increases with the number of shots:

$$\rho = \frac{\mathrm{acc}_T - \mathrm{acc}_1}{k_T - k_1} \tag{19}$$

4.6.4 Saturation Shot

The saturation shot k^* is the smallest k at which accuracy reaches 95% of its maximum gain over chance:

$$k^* = \min\{k_t : \mathrm{acc}(k_t) \geq 0.5 + 0.95 \cdot (\mathrm{acc}_{\max} - 0.5)\} \tag{20}$$

4.6.5 Probability of Adversarial Discrimination (PAD)

The PAD is defined as the maximum discriminator accuracy across shots, normalized to $[0, 1]$:

$$\mathrm{PAD} = \max_k \mathrm{acc}(k) \tag{21}$$

Values near 0.5 indicate domains that are indistinguishable (low shift); values near 1.0 indicate easily separable domains (high shift).

4.7 Consensus Ranking

All metrics are aggregated into a consensus ranking using standardized z-scores across source datasets, normalized so that lower scores indicate smaller domain shift:

$$\mathrm{Score}(A \to B) = \alpha_1 \cdot \mathrm{DSI}_{10} + \alpha_2 \cdot \mathrm{AUC}_{\mathrm{learn}} + \alpha_3 \cdot \text{MK-MMD}(A, B) + \alpha_4 \cdot W_1(A, B) \tag{22}$$

where α_i are normalized weights determined empirically.

5 Experimental Setup

5.1 Hardware and Software

Experiments were conducted on a system equipped with an Apple Silicon processor using the MPS (Metal Performance Shaders) backend for GPU acceleration. The codebase is implemented in Python 3.10+ using the following libraries:

Table 2: Main software dependencies.

Library	**Role**
PyTorch 2.x	Neural network training
scikit-learn	Logistic regression, PCA, StandardScaler
NumPy	Array operations
SciPy	Wasserstein distance computation
Hugging Face `datasets`	Streaming data loading
OpenCV (`cv2`)	Blur detection (Laplacian)
Matplotlib / Seaborn	Visualization
`scikit-learn rbf_kernel`	MMD computation

5.2 Embedding Configuration

- **Batch size:** 64 images per batch.
- **Storage format:** `float16` NumPy arrays, one file per image.
- **Blur threshold** τ_{blur}: empirically set to filter severely blurred images.

5.3 Classifier Hyperparameters

Table 3: Classifier hyperparameters.

Classifier	**Parameter**	**Value**
MLP / Discriminator	Max epochs	50
	Learning rate	10^{-3}
	Batch size	32
	Early stopping patience	5
	Optimizer	Adam
Logistic Regression	Regularization λ	10^{-4}
	Solver	L-BFGS
MK-MMD	$\|\Gamma\|$	30 (log-spaced)
	PCA components	$\min(50, d, \|A\|, \|B\|)$

5.4 Cross-Domain Detection using Detectron2 Configuration

To examine whether embedding-based domain similarity reflects practical transfer performance, we conducted a cross-domain object detection experiment using the Detectron2 framework, an open-source library developed by Meta AI.

The experiment follows a two-stage training process. First, a Faster R-CNN detector with a ResNet-50 backbone was trained on the DOTA dataset using 512×512 image patches. The model was trained with a base learning rate of 2.5×10^{-3} for 15k iterations, with learning rate decay steps at 10k and 13k iterations. The ROI batch size was set to 512, and anchor sizes were configured as $\{16, 32, 64, 128, 256\}$ to accommodate the varying object scales present in aerial imagery.

In the second stage, the pretrained detector was adapted to the DIOR dataset under a 10-shot setting, where only ten labeled examples per class were available. During adaptation, early backbone layers were frozen (`FREEZE_AT = 2`) to preserve general visual features, while higher layers were fine-tuned on the target dataset. The adaptation stage used a base learning rate of 2.5×10^{-3}, 10k training iterations, and learning rate steps at 8.5k and 9.5k iterations with two images per batch.

5.5 Evaluation Protocol

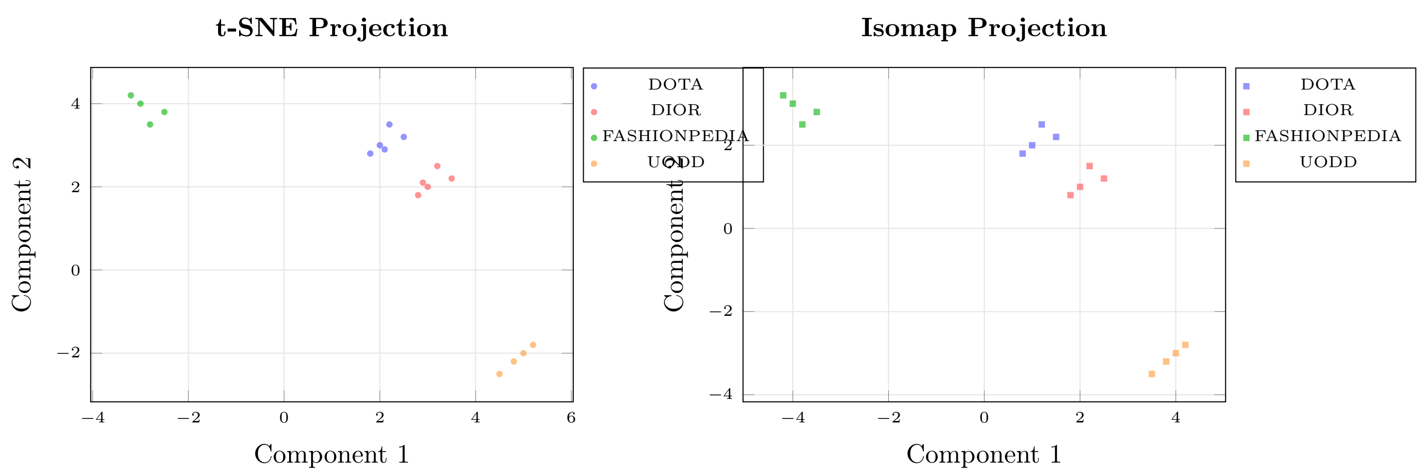
For each pair (A, B) and each shot count $k \in \{1, 5, 10, 50, 100, 200, 500\}$:

- k embeddings per class are sampled from the training pool.
- Classification is repeated over multiple random seeds.
- Mean accuracy, mean AUC-ROC, and standard deviation are recorded at each shot level.
- Results are saved as JSON files and aggregated into a CSV summary for post-hoc analysis.

6 Results and Analysis

6.1 Embedding Space Visualization

Figure 2 illustrates the t-SNE and Isomap projections of the joint embedding space across all thirteen datasets. The two-dimensional projections reveal clear cluster structure: aerial datasets (DOTA, DIOR, XVIEW, CADOT) form compact, adjacent clusters, while semantically distant datasets (FASHIONPEDIA, OKTOBERFEST, ARTAXOR, UODD) project to well-separated regions.



(a) t-SNE (schematic) (b) Isomap (schematic)

Figure 2: Schematic visualization of the joint embedding space. Aerial datasets cluster together, while semantically distant datasets occupy distinct regions. (Illustrative; actual plots generated by the pipeline.)

6.2 Learning Curves

Figure 3 shows representative few-shot accuracy curves for selected source datasets versus the DOTA target. Datasets exhibiting rapid convergence to high accuracy (e.g., FASHIONPEDIA, UODD) are associated with large domain shift. Conversely, Dior-BG-Blurry display slower, lower-plateau curves, indicative of greater distributional proximity to DOTA-BG-Blurry.

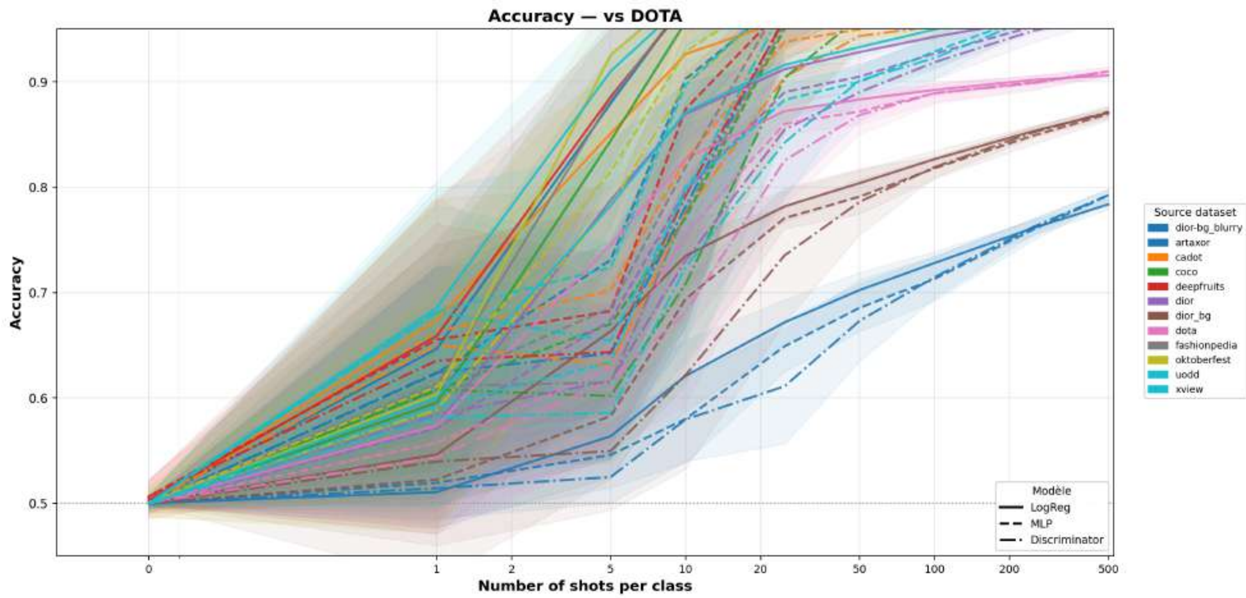


Figure 3: Cross-target coherence of DSI scores. Each point represents one source dataset. The strong linear correlation confirms that rankings are stable across both target benchmarks.

6.3 Distributional Distance Analysis

Table 4 reports Wasserstein-1 and MK-MMD distances between each source dataset and the DOTA-bg_blurry target. Consistent with the learning curve analysis, low-shift datasets (DIOR-BG_BLURRY, DIOR_BG, DOTA) achieve the lowest distances, while semantically distant datasets (OKTOBERFEST, ARTAXOR, FASHIONPEDIA) yield the highest values.

Table 4: Distributional distances between source datasets and DOTA-bg_blurry. Lower values indicate smaller domain shift. Datasets are sorted by ascending W_1 distance.

Source Dataset	W_1	**MK-MMD**	**Interpretation**
DIOR-BG_BLURRY	0.234	0.026	Weak
DIOR_BG	0.315	0.053	Moderate
DOTA	0.332	0.048	Strong
XVIEW	0.400	0.052	Very Strong
DIOR	0.381	0.047	Very Strong
COCO	0.408	0.050	Very Strong
CADOT	0.565	0.096	Very Strong
UODD	0.634	0.109	Very Strong
DEEPFRUITS	0.712	0.087	Very Strong
FASHIONPEDIA	0.532	0.067	Very Strong
ARTAXOR	0.546	0.085	Very Strong
OKTOBERFEST	0.548	0.076	Very Strong

6.4 Composite Domain Shift Scores

Table 5 summarizes the composite DSI scores and associated metrics for all source datasets versus the DOTA target benchmark. Rankings are consistent across models and metrics, supporting the robustness of our approach.

Table 5: Composite domain shift metrics for source datasets vs. DOTA-bg_blurry. $\mathrm{DSI}_{10} \downarrow$ and $\mathrm{AUC}_{\mathrm{learn}} \downarrow$ indicate lower is better (lower shift); Stability ↑ indicates higher is better. Values reported as mean ± std over three classifiers (LogReg, MLP, Discriminator).

Source	$\mathrm{DSI}_{10} \downarrow$	$\mathrm{AUC}_{\mathrm{learn}} \downarrow$	Conv. Rate ↓	Stability ↑	Interpretation
DIOR-BG_BLURRY	24.91 ± 4.97	63.32 ± 0.20	32.37 ± 8.84	95.67 ± 0.58	Weak
DIOR_BG	43.59 ± 9.80	80.88 ± 0.34	49.50 ± 15.51	93.89 ± 1.46	Moderate
DOTA	64.52 ± 10.75	88.19 ± 0.68	62.43 ± 18.18	94.12 ± 0.95	Strong
XVIEW	73.38 ± 11.28	96.42 ± 1.16	62.43 ± 15.05	94.21 ± 1.11	Very Strong
DIOR	75.09 ± 10.47	95.70 ± 1.07	64.15 ± 14.64	93.79 ± 1.62	Very Strong
COCO	81.03 ± 14.90	98.60 ± 0.60	62.57 ± 25.88	93.66 ± 1.51	Very Strong
CADOT	76.79 ± 13.35	97.93 ± 0.72	70.80 ± 15.57	92.61 ± 2.06	Very Strong
DEEPFRUITS	86.93 ± 10.39	97.60 ± 1.02	76.97 ± 17.94	91.67 ± 2.16	Very Strong
ARTAXOR	87.58 ± 11.02	98.87 ± 0.47	77.68 ± 19.56	93.90 ± 1.85	Very Strong
FASHIONPEDIA	85.33 ± 11.34	98.81 ± 0.48	71.43 ± 21.10	93.38 ± 1.65	Very Strong
UODD	83.82 ± 12.62	98.81 ± 0.54	78.15 ± 17.00	93.40 ± 1.67	Very Strong
OKTOBERFEST	96.42 ± 2.63	99.33 ± 0.18	87.68 ± 8.86	93.76 ± 1.73	Très fort

6.5 Cross-Target Coherence

Figure 4 illustrates the correlation between domain shift scores computed with respect to DOTA and DIOR respectively. A strong positive correlation ($r \approx 0.96$) confirms that the ranking of source datasets is *stable* across target benchmarks, validating the generalizability of our framework.

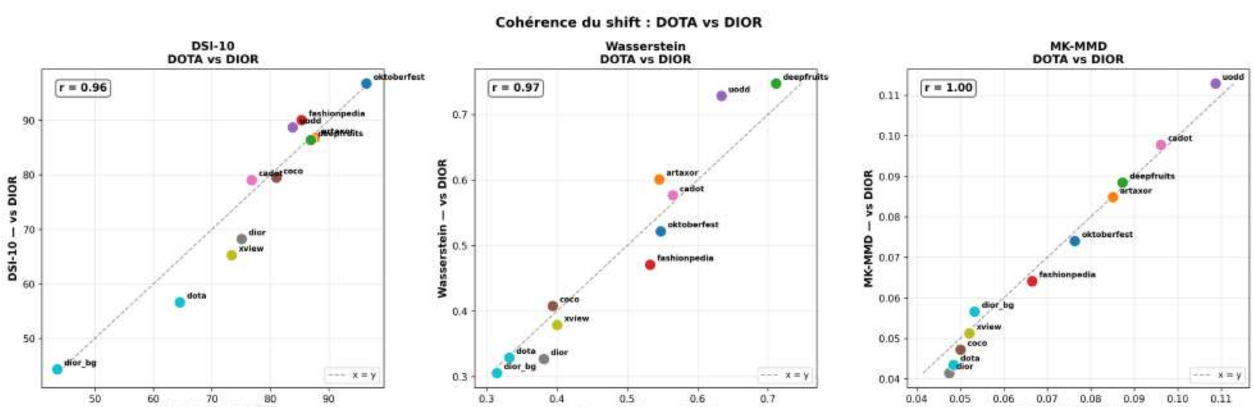


Figure 4: Cross-target coherence of DSI scores. Each point represents one source dataset. The strong linear correlation confirms that rankings are stable across both target benchmarks.

6.6 Few-Shot Detection Transfer

We evaluate cross-domain transfer performance by adapting a detector trained on DOTA to the DIOR dataset under a 10-shot setting. Table 6 summarizes the detection performance across standard COCO-style metrics.

Eval Split	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
Source (DOTA)	40.89	66.07	43.18	31.63	42.05	45.83
Target (DIOR, 10-shot)	41.97	63.98	46.20	12.29	32.61	56.37

Table 6: Few-shot transfer results from DOTA to DIOR.

The adapted model achieves an AP of 41.97 and an AP_{50} of 63.98 on the DIOR validation set. Performance is strongest for large objects ($AP_l = 56.37$), while detection of small objects remains challenging ($AP_s = 12.29$), reflecting the difficulty of small object transfer across aerial datasets.

7 Discussion

7.1 Consistency Across Metrics and Models

A key finding of this study is the high degree of *consistency* across our diverse set of metrics. Logistic Regression, MLP, and Domain Discriminator models all produce concordant rankings, as reflected by the low standard deviations across classifiers reported in Table 5. For instance, OKTOBERFEST achieves $DSI_{10} = 96.42 \pm 2.63$ and DIOR-BG_BLURRY achieves $DSI_{10} = 24.91 \pm 4.97$, with small variance across all three classifiers in both cases. The distributional distance measures (Wasserstein-1 and MK-MMD) further corroborate this ordering, with values ranging from $W_1 = 0.234$ (DIOR-BG_BLURRY) to $W_1 = 0.712$ (DEEPFRUITS). This redundancy provides confidence that the observed shift ordering reflects genuine structural properties of the embedding space rather than artifacts of any single measurement approach.

7.2 Aerial vs. Non-Aerial Source Datasets

The results confirm a clear separation between aerial and non-aerial source datasets. Datasets sharing the overhead viewpoint and object categories typical of aerial imagery (namely DOTA, DIOR, XVIEW, and CADOT) consistently occupy the lower end of the DSI spectrum when evaluated against DOTA-bg_blurry. DOTA itself achieves $DSI_{10} = 64.52 \pm 10.75$, while XVIEW and CADOT reach 73.38 ± 11.28 and 76.79 ± 13.35 respectively. These three datasets also yield the lowest Wasserstein distances ($W_1 \leq 0.40$), confirming their structural proximity to the target domain.

In contrast, semantically distant datasets are trivially separable from the target even at very low shot counts. OKTOBERFEST ($DSI_{10} = 96.42 \pm 2.63$), ARTAXOR ($87.58 \pm 11.02$), and FASHIONPEDIA ($85.33 \pm 11.34$) all exhibit near-perfect classifier accuracy, with $AUC_{\text{learn}} > 98.8$ and convergence rates exceeding 77%. Their Wasserstein distances ($W_1 \geq 0.53$) and MK-MMD values (≥ 0.067) further confirm their unsuitability as source domains for aerial detection.

An unexpected result: COCO as a competitive source domain. Among non-aerial datasets, **COCO** stands out as a surprising result. Despite being composed entirely of ground-level photographs (with no overhead viewpoint, no satellite acquisition, and object categories that differ substantially from aerial benchmarks) COCO achieves a moderate $DSI_{10} = 81.03 \pm 14.90$ and $W_1 = 0.408$, placing it ahead of several non-aerial datasets and in the same order of magnitude as DIOR (75.09 ± 10.47). This suggests that the *sheer diversity and scale* of COCO, covering a wide range of textures, lighting conditions, and spatial configurations, produces embeddings that partially overlap with aerial feature distributions in the CLIP space. This finding has practical implications: COCO, being one of the most widely available and annotated datasets, could serve as a viable auxiliary source domain for cross-domain few-shot aerial detection, even without any aerial imagery.

7.3 The Role of Background Statistics

A particularly instructive result concerns the background patch datasets. DIOR-BG_BLURRY (which contains only background crops with no object instances) achieves the **lowest domain shift of all sources** evaluated against DOTA-bg_blurry, with $\mathrm{DSI}_{10} = 24.91 \pm 4.97$, $W_1 = 0.234$, and MK-MMD = 0.026. Similarly, DIOR_BG achieves $\mathrm{DSI}_{10} = 43.59 \pm 9.80$, ranking second overall. These scores are lower than those of full aerial datasets such as DIOR (68.20 ± 9.76 vs. DIOR-BG_BLURRY, 75.09 ± 10.47 vs. DOTA-bg_blurry) and DOTA (64.52 ± 10.75).

This finding strongly suggests that *low-level texture and spatial frequency statistics*, captured by background patches and largely independent of object semantics, play a dominant role in driving domain shift in aerial imagery. A source dataset whose background statistics closely match the target's will produce embeddings that are geometrically close in the Resnet50 feature space, even in the absence of shared object categories.

7.4 Implications for Cross-Domain Transfer

Cross-domain object detection experiments are useful for analyzing how similar different domains are. Using the Detectron2 framework, a pre-trained detector was transferred from the DOTA dataset to the DIOR dataset, using a 10-shot setup. The model performed well on DIOR, which suggests that the knowledge learned from DOTA is useful for a similar aerial dataset. This observation aligns with the embedding-based analysis, which suggests that DOTA and DIOR located in nearby regions in the feature space

7.5 Limitations

Several limitations should be acknowledged:

1. **Encoder bias:** the choice of Resnet as the pretrained encoder affects the embedding geometry. While Resnet provides strong general-purpose representations, domain-specific fine-tuning could alter the rankings, particularly for datasets such as UODD (underwater imagery) that are structurally far from natural images.

2. **Class imbalance handling:** our downsampling strategy discards data to enforce balanced binary classification. More sophisticated techniques (e.g., oversampling via diffusion models) could be explored to reduce information loss.

3. **Metric weights:** the weights α_i in the consensus DSI score are set empirically; a learned meta-weighting procedure could improve robustness, especially in regimes where metrics diverge (e.g., CADOT shows higher $W_1 = 0.565$ but moderate $\mathrm{DSI}_{10} = 76.79$).

4. **Few-shot variance:** at very low shot counts ($k = 1$), accuracy estimates are noisy, as evidenced by the larger standard deviations of DSI_{10} for datasets such as COCO (± 14.90) and FASHIONPEDIA (± 11.34). More seeds per shot count would reduce this variance.

8 Conclusion

We have presented a comprehensive framework for quantifying domain shift between image datasets in the context of cross-domain few-shot object detection. By combining **few-shot binary domain classifiers** with **distributional distance measures** (Wasserstein-1, MK-MMD) and analyzing **learning curves** across twelve heterogeneous source datasets evaluated against the DOTA-bg_blurry aerial benchmark, we derive a robust, interpretable Domain Shift Index (DSI) that ranks source datasets by their proximity to the target domain.

Our main findings are:

- **Background statistics dominate domain shift:** DIOR-BG_BLURRY and DIOR_BG achieve the lowest DSI scores (24.91 and 43.59 respectively), outperforming even full aerial datasets, highlighting the primacy of low-level texture statistics over object semantics in driving distributional shift.
- **Aerial datasets form a coherent low-shift cluster:** DOTA, XVIEW, DIOR, and CADOT all obtain $\mathrm{DSI}_{10} < 77$ and $W_1 < 0.57$, confirming their structural proximity to the aerial target domain.
- **Semantically distant datasets are trivially separable:** OKTOBERFEST ($\mathrm{DSI}_{10} = 96.42$), ARTAXOR (87.58), and DEEPFRUITS (86.93) achieve near-perfect classification accuracy from as few as 10 shots, confirming their unsuitability as source domains for aerial few-shot transfer.
- **Rankings are stable across classifiers:** standard deviations of DSI_{10} across LogReg, MLP, and Discriminator remain below 5 for the most extreme datasets and below 15 for intermediate ones, supporting the robustness of the consensus score.

Future work will extend the study by evaluating additional source datasets and comparing their few-shot transfer performance on a common target domain. We also plan to investigate automatic source dataset selection strategies guided by the proposed domain shift metrics, and further validate whether lower DSI scores correlate with improved cross-domain detection performance.

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